

with marketing. For this position, you should be a B.Tech or M.Tech, and have the basic knowledge of solar systems, PV panels, inverter systems, DC cabling and SCADA systems. You should be highly analytic and capable of handling all the test equipment including logic analysers and remote monitoring. Knowledge of financial aspects of project report preparation, cost estimation and proposal preparation is required. Also, you must possess excellent communication skills, negotiation skills and analytical skills. You must be able to foresee the things and take actions proactively.

Process engineers are required to have experience in solar photovoltaic cell/module or semiconductor device manufacturing.

If you are interested in R&D, you should either have a Ph.D in solidstate physics, BE/B.Tech in electronics & communication engineering, or M.Sc. in electronics or solidstate physics. For a senior position, some experience in electronics R&D, and technology management and product development processes is required.

To get into quality assurance, you should be a BE or B.Tech in electronics and communication engineering or in electronics and electrical engineering with experience in the field of quality management in either a manufacturing, electronics, light engineering or semiconductor industry. You should be well versed with advanced practices and concepts in quality management and have the capacity to drive quality culture across all functional areas.

BE in electronics with MBA and two-three years of experience in industrial marketing in the solar/renewable energy field will qualify you for the

Solar fundamentals

The basic components of a solar power system are:

- A set of PV panels that transform sunlight (photons) into DC electricity
- A racking system that firmly holds the panels to the roof and exposes the panels to the sun at an advantageous angle
- Inverters that transform DC electricity into AC electricity
- Wiring that connects everything
- Storage battery (in the case of grid-fed power plant, big-size battery is not necessary to store and use power after sunset)

Solar cells are the basic elements of a solar module or panel. These cells are made of silicon—a common semiconductor material.

The manufacturing base in India comprises cells and module manufacturing. For polysilicon or crystalline wafers—the key raw material for cell manufacturing—the country is dependent on imports. Indian companies enter the value chain from wafer onwards and manufacture PV cells from these and then assemble the cells into modules and so on.

position of sales and marketing engineer. You should be willing to travel to different places. As you gain experience, you can grow to the position of manager for sales and marketing.

For equipment maintenance, you should be a BE or B.Tech in electronics and electrical engineering with experience in equipment maintenance in a manufacturing industry. You should have worked with programmable logic control based equipment. Also, you should be good at troubleshooting and running machines with minimal down time.

If aiming for production management, you should be a BE or B.Tech in electronics/instrumentation/mechatronics or M.Sc. in physics/electronics with experience in a manufacturing environment—preferably in the electronics or light engineering industry. Candidates with experience in semiconductor devices, solar photovoltaic cells or modules manufacturing are preferred. A good insight of modern manufacturing practices and prior work experience in a systems based environment are desirable.

For project management, you

should have experience in turnkey project management in a manufacturing industry. Prior experience in handling major projects under challenging conditions is desirable.

Where to study

The solar PV industry is in a very nascent stage and the infrastructure not completely built in India. However, with the rapid development of this industry, there will be requirement of fresh graduates and postgraduates who can be trained in the respective areas.

There is a huge talent crunch as few colleges or institutes offer courses dedicated to solar energy or renewable energy. However, there are about 25 institutes in India offering courses in energy studies, with renewable energy as one of the subjects.

Most of the courses in renewable energy are available at the master's level. Some of the institutes and universities offering these courses are Department of Energy Science & Engineering, IIT Mumbai; Center of Energy Studies, IIT, New Delhi; School of Energy Studies, Jadavpur University, Kolkata; Department of School of Energy Studies, University of Pune; TERI University, New Delhi; University of Lucknow; and Pandit Deendayal Petroleum University (Gandhinagar, Gujarat). Outside India, University of Dundee in Scotland and University of Nottingham in England also offer this course.

Also, there are centres of energy

To fulfil the target of 20,000 MW of installed solar power capacity under the National Solar Mission (NSM), the Indian solar sector would need over 100,000 skilled solar professionals. Considering the potential employment generation by state solar policies, the cumulative figure would be much higher